

WISP.data: An open-source R package for processing spectral data from above-water autonomous spectroradiometers

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Summary

WISP.data is an R package designed to automate the download, quality control (QC), processing, analysis, and visualization of spectral data collected by WISPstation fixed spectroradiometers. Developed by Water Insight B.V. (Ede, The Netherlands) as an evolution of the handheld WISP-3 system (Hommersom et al., 2012), the WISPstation is an autonomous above water instrument installed at fixed position that records radiance and irradiance across a wavelength range of 350 to 900 nm, with a spectral resolution of 4.6 nm at high frequency (e.g. every 15 minutes, during daytime) (Peters et al., 2018). This R package serves as a crucial bridge between Water Insight API services and the scientific users, enabling the conversion of native reflectance measurements into robust, analysis-ready products through fully reproducible and transparent workflows, from an Open Science perspective.

WISP.data provides a modular set of R functions for: I. downloading data from specific WISPstation by making use of the solutions provided by Water Insight itself (APIs and now via user accounts); II. performing automated and reproducible quality control (QC); III. performing automated sky-glint removal (SR); IV. applying empirical and semi-analytical bio-optical algorithms to spectral measurements to retrieve optical properties and water quality parameters; V. visualizing spectral and derived data products.

This makes WISP.data an ideal solution for satellite-derived product validation, operational water quality monitoring, long-term research applications, and integration into large-scale environmental data pipelines.

Statement of need

The WISPstation is an autonomous above water fixed spectrometer installed at a fixed position, which supports continuous monitoring of water quality; beyond providing high-frequency spectral measurements, it provides water quality products derived from selected algorithms (Gons et al., 1997, 2005; Simis et al., 2005), essential for environmental observation, ecosystem assessment, and long-term trend analysis.

The WISPstation operates with 8 specialized channels to optimize data collection:

- Dual-directional radiance:** two sets of sensors (looking NNW and NNE) measure upwelling

radiance (L_u) and sky radiance (L_{sky}) at 40° angles.

- **Irradiance and calibration:** includes two downwelling irradiance (E_s) channels and two unexposed “dark” channels to assess sensor degradation over time.

The system automatically selects the best-oriented sensor set based on the Sun’s position, maintaining a relative azimuth angle of approximately 135°.

A central function of WISP.data is the retrieval and management of Remote Sensing Reflectance (R_{rs}), which the WISPstation calculates as the ratio between water-leaving radiance (L_w) and downwelling irradiance (E_s) (Mobley, 1999):

$$R_{rs}(\lambda) = \frac{L_w(\lambda)}{E_s(\lambda)} = \frac{L_u(\lambda) - \rho \cdot L_{sky}(\lambda)}{E_s(\lambda)}$$

Where ρ is the Fresnel reflection coefficient, L_u is the total upwelling radiance and L_{sky} is the sky radiance contributing to surface reflections.

However, the effective management and scientific use of these spectral data and derived products present several significant challenges. Data retrieval through API services is often labor-intensive and technically demanding, especially for users without advanced programming experience. In addition, native spectral measurements could be affected by radiometric issues making the implementation of consistent and rigorous quality control procedures essential. Without rigorous filtering and validation protocols, derived products can be unreliable or scientifically misleading. A further critical barrier lies in the interpretation of spectral signatures themselves. For non-expert users, it is difficult to assess the physical and optical reliability of reflectance spectra, identify anomalous signals, or distinguish between instrument geometry artifacts and real environmental variability. Moreover, the application of third-party bio-optical algorithms for the estimation of water quality parameters typically requires substantial domain knowledge, careful parameterization, and consistent preprocessing workflows, which are rarely standardized across studies.

The scientific validity and operational reliability of WISPstation measurements have been demonstrated in several studies, ranging from the detection of climate-driven chlorophyll-a changes during extreme events (Free et al., 2021) to the analysis of phytoplankton spatio-temporal dynamics in Lake Trasimeno (Bresciani et al., 2020). Despite these successful applications, the processing of WISPstation data has been labor-intensive and time-consuming. Prior to the development of WISP.data, researchers often had to manually inspect individual spectral signatures to identify outliers before the data could be used to estimate water quality parameter. This manual quality control process is prone to subjectivity and significantly limits the scalability of high-frequency monitoring.

WISP.data addresses these challenges by delivering an integrated, transparent, reproducible, and user-oriented software ecosystem that unifies data acquisition, quality control, and product generation within a single R-based framework. By lowering technical and methodological barriers, the package enables both expert and non-expert users to transform native WISPstation measurements into reliable, scientifically consistent water quality products, fostering reproducibility, comparability, and broader adoption of spectral monitoring technologies in aquatic research and operational monitoring.

State of the field

The current benchmark for processing WISPstation data is WISPcloud (<https://wispcloud.waterinsight.nl/>) a proprietary web-based infrastructure developed by Water Insight. While WISPcloud provides an efficient ‘plug-and-play’ solution by delivering ready-to-use products, its closed proprietary architecture poses significant challenges for rigorous scientific inquiry. Specifically, the platform

lacks the flexibility to modify critical processing parameters, test third-party algorithms, or customize site-specific quality control thresholds.

In the R ecosystem, while established packages exist for general hyperspectral analysis (e.g., hyperSpec (Beleites et al., 2024)) or for specific in-water optical profilers (e.g., cops (Rusch et al., 2021; Rusch & Mair, 2024)), there is a notable software gap for autonomous, fixed, above-water spectroradiometers. Rather than merely replicating existing cloud functionalities, WISP.data transforms a proprietary workflow into an open, verifiable, and scientifically controllable ecosystem.

Software design

The choice of R as the development environment is motivated by its capacity for developing well-curated, extensively documented packages within a coherent methodological structure. This choice is further justified by R's high effectiveness in statistical data analysis—a central aspect of this study—and its native support for reproducible scientific workflows, which aligns with the Open Science goals of WISP.data.

The design philosophy of WISP.data is based on three characteristics: I. a modular, pipeline-oriented workflow abstracting complex API interactions; II. the adoption of community standards for data manipulation and physical units; III. a “transparent-box” approach to quality control, sky-glint removal, and bio-optical algorithms application. The complete workflow of the WISP.data package is shown in Figure 1.

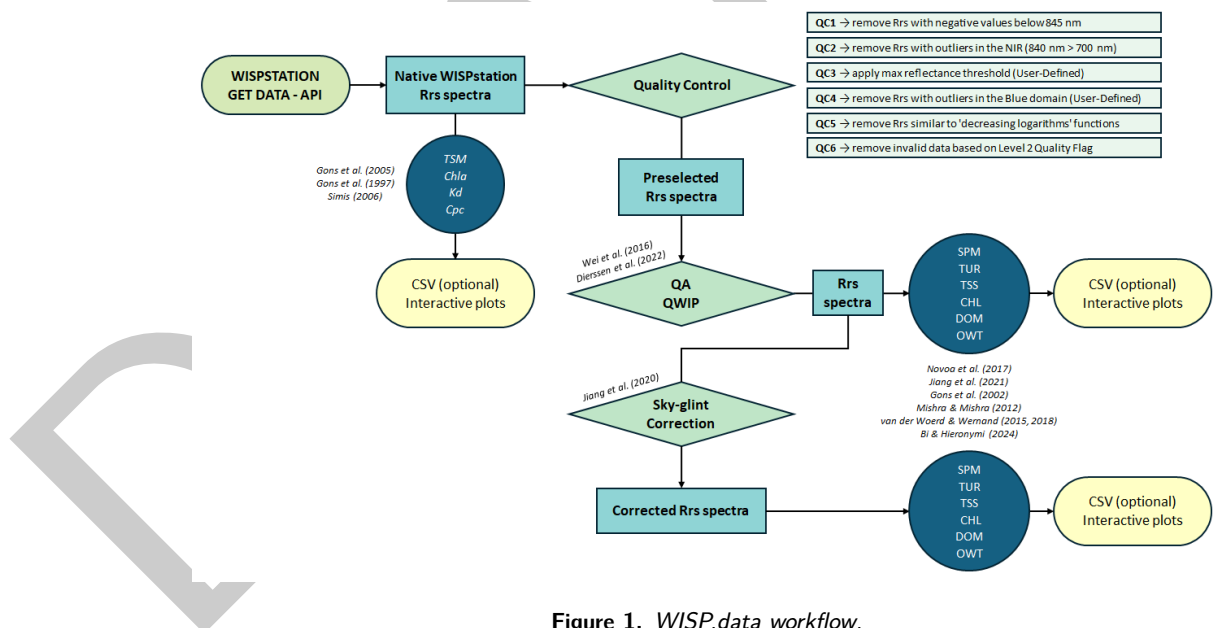


Figure 1. WISP.data workflow.

The package architecture is organized into functional modules:

- **Data Integration Layer:** the functions `wisp_get_reflectance_data()` and its iterative wrapper `wisp_get_reflectance_multi_data()` manage the communication with the Water Insight API using the `httr2` framework. These functions allow users to filter measurements by specific dates and time periods. Raw JSON/text responses are parsed, unnested, and converted into structured tibble objects. This layer also handles the transformation of spectral columns into a standardized format, making the data immediately ready for analysis.

- 111 ■ **Physical Awareness:** WISP.data integrates the units package to maintain physical
112 consistency. Every measurement retrieved is assigned its proper unit, preventing
113 dimensional errors in downstream calculations.
- 114 ■ **Modular Quality Control (QC):** the `wisp_qc_reflectance_data()` function implements
115 a multi-step diagnostic pipeline, allowing for a sequential execution of six distinct QC
116 tests (QC1–QC6) alongside advanced validation metrics like Quality Assurance (QA)
117 (Wei et al., 2016) and Quality Water Index Polynomial (QWIP) (Dierssen et al., 2022).
118 The architecture is highly parameterized, enabling researchers to tune thresholds (e.g.,
119 `maxPeak`, `maxPeak_blue`, `qa_threshold`, and `qwip_threshold`) to suit specific optical
120 water types, from oligotrophic to highly turbid and productive environments.
- 121 ■ **Sky-glint removal (SR):** the `wisp_sr_reflectance_data()` function implements a sky-
122 glint removal algorithm based on the methodology proposed by Jiang et al. (2020).
- 123 ■ **Application of third-party algorithms:** WISP.data currently supports a wide array of
124 standardized algorithms for assessing the physical and optical properties of water (Bi
125 & Hieronymi, 2024; Lee et al., 2002; van der Woerd & Wernand, 2015, 2018) and for
126 estimating the concentrations of water quality parameters (Gons, 1999; Gons et al.,
127 2002; Jiang et al., 2021; Mishra & Mishra, 2012; Nechad et al., 2010; Novoa et al.,
128 2017). These are implemented as modular flags within `wisp_qc_reflectance_data()`
129 and `wisp_sr_reflectance_data()` functions, allowing for a seamless “one-call” workflow
130 from native spectra to validated products.
- 131 ■ **Visualization layer:** utilizing ggplot2 and plotly as backends, the package provides
132 consistent visualization interfaces. Functions like `wisp_plot_reflectance_data()`,
133 `wisp_plot_comparison()` (Figure 2), and `wisp_trend_plot()` (Figures 3, 4) are
134 designed to handle both publication-quality plots and interactive exploration.

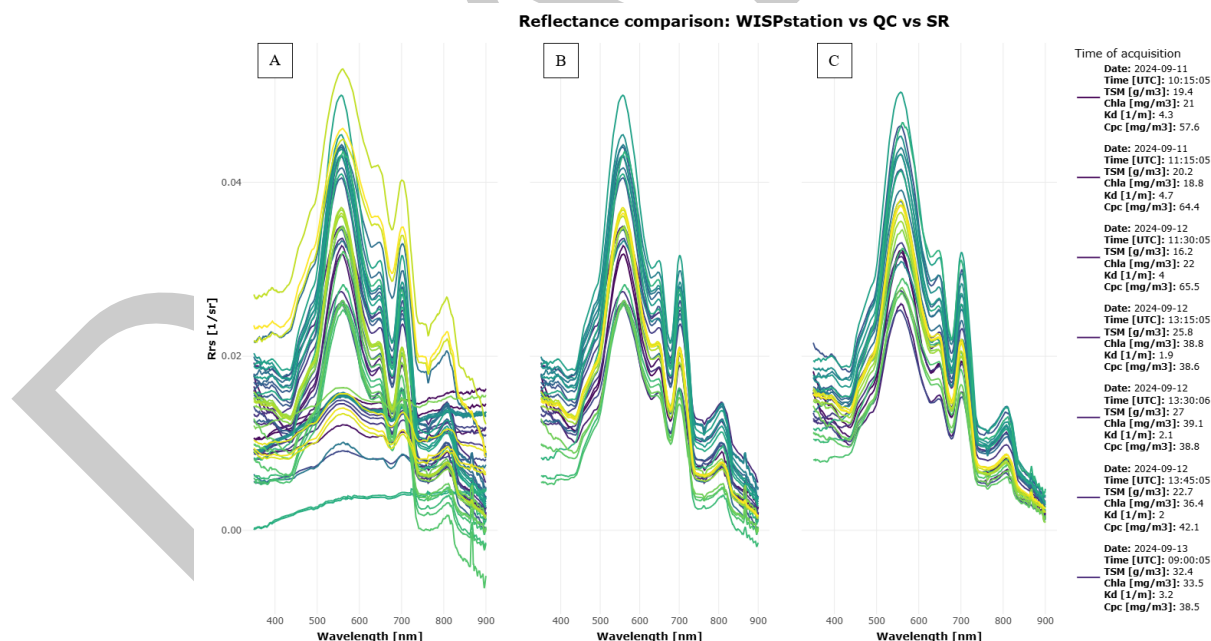


Figure 2. Plot resulting from the `wisp_plot_comparison()` function showing the comparison between:
A) native WISPstation R_{rs} , B) R_{rs} filtered by “QC”, C) R_{rs} to which “SR” has been applied (site:
Trasimeno; period: 11/09/2024 – 17/09/2024).

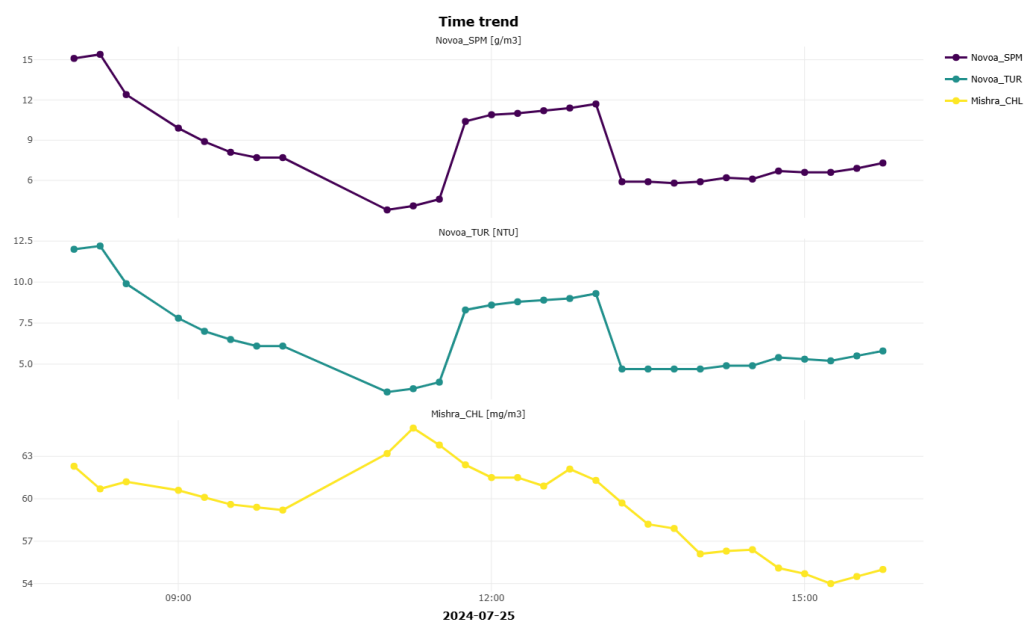


Figure 3. Plot resulting from the `wisp_trend_plot()` function showing the temporal trend of three exemplary parameters (from top to bottom: “Novoa_SPM”, “Novoa_TUR”, and “Mishra_CHL”) for 25/07/2024 from 8 a.m. to 4 p.m. (site: Trasimeno).

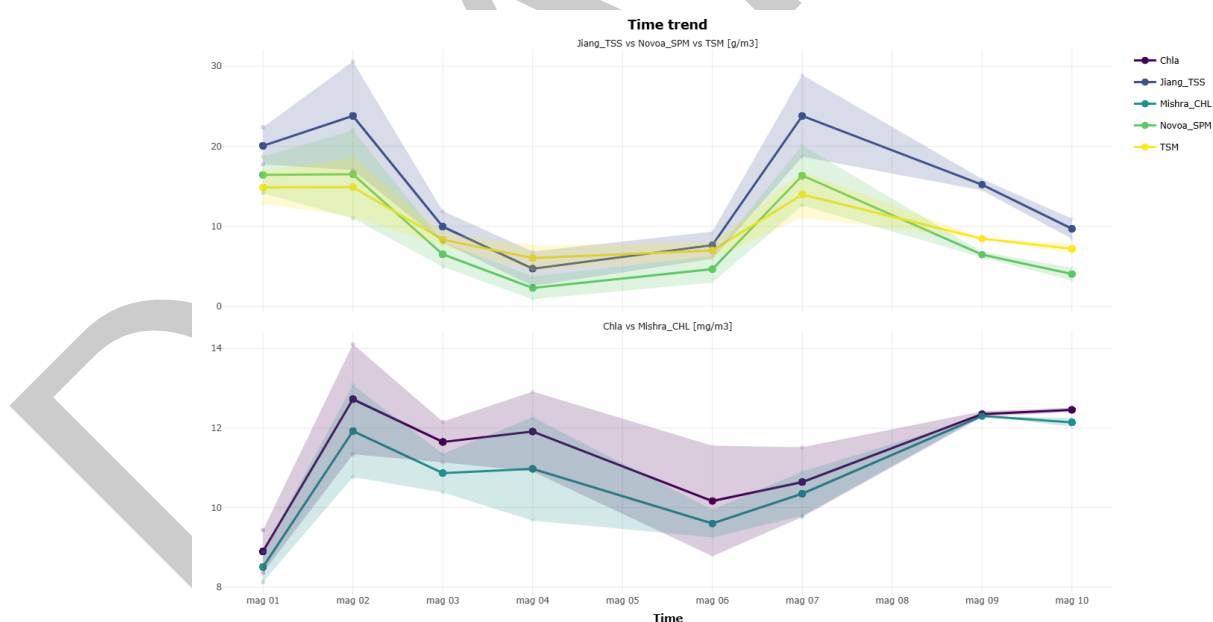


Figure 4. Plot resulting from the `wisp_trend_plot()` function showing the temporal trend of five exemplary parameters averaged on a daily basis: in the upper plot, a comparison between the WISPstation native algorithm (TSM) and two third-party algorithms for estimating suspended solids concentration (“Novoa_SPM” and “Jiang_TSS”); in the lower plot, the comparison between the WISPstation native chlorophyll-a algorithm (Chla) and Mishra_CHL algorithm. (site: Trasimeno; period: 01/05/2024 – 10/05/2024).

135 Research impact statement

136 The package was successfully tested using two WISPstations installed in two different aquatic
137 environments in Italy. The first was installed in April 2018 in Lake Trasimeno, on a lake
138 platform located approximately 400 m from Polvese Island (43.122° N, 12.134° E). The second
139 was deployed in September 2024 along the Po River at Pontelagoscuro (44.889° N, 11.6106°
140 E), approximately 95 km upstream from its confluence with the northern Adriatic Sea, on a
141 floating monitoring platform operated by the Agenzia Interregionale per il fiume Po (AIPO).

142 Lake Trasimeno, located in central Italy, is the fourth-largest lake in the country (124 km²).
143 Its catchment is characterized by intensive agricultural practices and livestock farming, which
144 lead to significant nutrient loading. These conditions frequently promote phytoplankton
145 blooms, including potentially harmful cyanobacteria. The lake exhibits complex hydrodynamic
146 behavior, with strong intra-day and inter-day variability driven by meteorological forcing such as
147 temperature fluctuations and wind-induced sediment resuspension. The Po River, the longest
148 river in Italy, flows across the Po Valley and governs the exchanges at the river-sea interface,
149 driving the transfer of sediments, nutrients, and organic matter toward the northern Adriatic
150 Sea and influencing coastal biogeochemistry and water quality. The river is characterized by
151 highly dynamic hydrological conditions, with discharge variability driven by seasonal cycles
152 as well as extreme events such as floods and prolonged drought periods. These conditions
153 strongly affect sediment transport and turbidity, resulting in rapid changes in water optical
154 properties.

155 Both environments are the focus of long-standing research efforts aimed at understanding the
156 spatio-temporal dynamics of parameters that can also be monitored through remote sensing.
157 The stations are officially designated as research sites within European Strategy Forum on
158 Research Infrastructures (ESFRI). Particularly, Lake Trasimeno¹ is part of the Integrated
159 European Long-Term Ecosystem, Critical Zone and Socio-Ecological Research (eLTER RI)²,
160 while the Po River is a Supersite of DANUBIUS-RI³, the International Centre for Advanced
161 Studies on River-Sea Systems, a pan-European distributed Research Infrastructure enabling
162 integrated studies of rivers and their catchments, transitional waters such as estuaries, deltas
163 and lagoons, and their adjacent coastal seas.

164 Within these RIs, the availability of continuous, comparable, and quality-controlled
165 measurements is essential to minimize uncertainties and ensure data reliability. Such
166 measurements, acquired through high-quality sensor systems and complemented by parameters
167 derived from robust algorithms, can be particularly valuable for the calibration and validation
168 (Cal/Val) of satellite-derived products. Within eLTER RI, these measurements contribute
169 to the definition of standardized variables (Zacharias et al., 2026), known as Standard
170 Observations (SOs), which integrate target variables, measurement methods, and protocols
171 to ensure harmonized data collection across sites at the continental scale. Among these,
172 the SO “Phenological traits – SOBIO_15” specifically addresses the use of remote sensing
173 approaches for monitoring vegetation phenological dynamics, including in aquatic ecosystems,
174 encompassing both inland standing waters (lakes) and running waters (rivers).

175 The scientific utility of WISP.data has recently been demonstrated by Ghirardi et al. (under
176 review), who investigated the spatio-temporal dynamics of suspended particulate matter and
177 chlorophyll-a in Lake Trasimeno using a synergistic approach. By combining high-frequency in
178 situ observations processed with WISP.data and multisensor satellite imagery (2019–2024),
179 the study validated R_{rs} products from 14 different satellite sensors. The strong agreement
180 between satellite-derived estimates and the WISP.data processed ground-truth enabled the
181 accurate retrieval of both water quality parameters. These results highlight the package’s
182 role in bridging the gap between local autonomous measurements and large-scale satellite

¹<https://deims.org/c93c97fb-196e-480a-87b0-817218cd7c24>

²<https://elter-ri.eu>

³<https://danubius-ri.eu>

183 monitoring, facilitating the study of rapid environmental changes in lake ecosystems. For the Po
 184 River, WISP.data is used to process high-frequency hyperspectral radiometric data collected at
 185 the Pontelagoscuro platform, including quality control, sky-glint correction, and the application
 186 of algorithms to derive turbidity from optical measurements. The resulting time series of
 187 above-water observations supports the assessment of hyperspectral and multispectral satellite
 188 missions, enabling the testing of different atmospheric correction algorithms. In addition, water
 189 turbidity derived from WISPstation measurements is compared with in situ data acquired by
 190 in-water turbidimeters, enabling consistency evaluation between radiometric-derived and direct
 191 observations. This integrated approach enhances the interpretation of turbidity dynamics in
 192 highly variable riverine systems.

193 Looking ahead, the modular architecture of WISP.data is well-suited to be extended and adapted
 194 to other optical view fixed spectroradiometric systems widely used in the scientific community
 195 for water body monitoring, thereby promoting greater interoperability and standardization in
 196 the processing of in situ optical data.

197 AI usage disclosure

198 No generative AI tools were used in the writing of this manuscript, or the preparation of
 199 supporting materials. Generative AI tools were used as a support for code development.

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211 References

- 212 Beleites, C., Bonifacio, A., Dahms, M., Egert, B., Fuller, S., Gegzna, V., Guliev, R., Hanson,
 213 B. A., Hermes, M., Kammer, M., Kiselev, R., Mellor, S., Oduniyi, E., & Sergo, V. (2024).
 214 *hyperSpec: a package to handle hyperspectral data sets in R*. <https://r-hyperspec.github.io/hyperSpec>
 215
- 216 Bi, S., & Hieronymi, M. (2024). Holistic optical water type classification for ocean, coastal,
 217 and inland waters. *Limnology and Oceanography*, 69(7), 1547–1561. <https://doi.org/10.1002/lno.12606>
 218
- 219 Bresciani, M., Pinardi, M., Free, G., Luciani, G., Ghebrehiwot, S., Laanen, M., Peters, S., Della
 220 Bella, V., Padula, R., & Giardino, C. (2020). The Use of Multisource Optical Sensors to
 221 Study Phytoplankton Spatio-Temporal Variation in a Shallow Turbid Lake. *Water*, 12(1),
 222 284. <https://doi.org/10.3390/w12010284>
- 223 Dierssen, H. M., Vandermeulen, R. A., Barnes, B. B., Castagna, A., Knaeps, E., &
 224 Vanhellemont, Q. (2022). QWIP: A quantitative metric for quality control of aquatic
 225 reflectance spectral shape using the apparent visible wavelength. *Frontiers in Remote*
 226 *Sensing*, 3, 869611. <https://doi.org/10.3389/frsen.2022.869611>

- Free, G., Bresciani, M., Pinardi, M., Giardino, C., Alikas, K., Kangro, K., Rõõm, E. I., Vaičiūtė, D., Bučas, M., Tiškus, E., Hommersom, A., & Peters, S. (2021). Detecting Climate Driven Changes in Chlorophyll-a Using High Frequency Monitoring: The Impact of the 2019 European Heatwave in Three Contrasting Aquatic Systems. *Sensors*, 21(18), 6242. <https://doi.org/10.3390/s21186242>
- Gons, H. J. (1999). Optical teledetection of chlorophyll a in turbid inland waters. *Environmental Science & Technology*, 33(7), 1127–1132. <https://doi.org/10.1021/es9809657>
- Gons, H. J., Ebert, J., & Kromkamp, J. (1997). Optical teledetection of the vertical attenuation coefficient for downward quantum irradiance of photosynthetically available radiation in turbid inland waters. *Aquatic Ecology*, 31(3), 299–311. <https://doi.org/10.1023/A:1009902627476>
- Gons, H. J., Rijkeboer, M., & Ruddick, K. G. (2002). A chlorophyll-retrieval algorithm for satellite imagery (Medium Resolution Imaging Spectrometer) of inland and coastal waters. *Journal of Plankton Research*, 24(9), 947–951. <https://doi.org/10.1093/plankt/24.9.947>
- Gons, H. J., Rijkeboer, M., & Ruddick, K. G. (2005). Effect of a waveband shift on chlorophyll retrieval from MERIS imagery of inland and coastal waters. *Journal of Plankton Research*, 27(1), 125–127. <https://doi.org/10.1093/plankt/fbh151>
- Hommersom, A., Kratzer, S., Laanen, M. L., Ansko, I., Ligi, M., Bresciani, M., & Peters, S. W. M. (2012). Intercomparison in the field between the new WISP-3 and other radiometers (TriOS Ramses, ASD FieldSpec, and TACCS). *Journal of Applied Remote Sensing*, 6(1), 063615. <https://doi.org/10.1117/1.JRS.6.063615>
- Jiang, D., Matsushita, B., Pahlevan, N., Gurlin, D., Lehmann, M. K., Fichot, C. G., & O'Donnell, D. (2021). Remotely estimating total suspended solids concentration in clear to extremely turbid waters using a novel semi-analytical method. *Remote Sensing of Environment*, 258, 112386. <https://doi.org/10.1016/j.rse.2021.112386>
- Jiang, D., Matsushita, B., & Yang, W. (2020). A simple and effective method for removing residual reflected skylight in above-water remote sensing reflectance measurements. *ISPRS Journal of Photogrammetry and Remote Sensing*, 165, 16–27. <https://doi.org/10.1016/j.isprsjprs.2020.05.003>
- Lee, Z., Carder, K. L., & Arnone, R. A. (2002). Deriving inherent optical properties from water color: a multiband quasi-analytical algorithm for optically deep waters. *Applied Optics*, 41(27), 5755–5772. <https://doi.org/10.1364/AO.41.005755>
- Mishra, S., & Mishra, D. R. (2012). Normalized difference chlorophyll index: A novel model for remote estimation of chlorophyll-a concentration in turbid productive waters. *Remote Sensing of Environment*, 117, 394–406. <https://doi.org/10.1016/j.rse.2011.10.016>
- Mobley, C. D. (1999). Estimation of the remote-sensing reflectance from above-surface measurements. *Applied Optics*, 38(36), 7442–7455. <https://doi.org/10.1364/AO.38.007442>
- Nechad, B., Ruddick, K. G., & Park, Y. (2010). Calibration and validation of a generic multisensor algorithm for mapping of total suspended matter in turbid waters. *Remote Sensing of Environment*, 114(4), 854–866. <https://doi.org/10.1016/j.rse.2009.11.022>
- Novoa, S., Doxaran, D., Ody, A., Vanhellemont, Q., Lafon, V., Lubac, B., & Gernez, P. (2017). Atmospheric corrections and multi-conditional algorithm for multi-sensor remote sensing of suspended particulate matter in low-to-high turbidity levels coastal waters. *Remote Sensing*, 9(1), 61. <https://doi.org/10.3390/rs9010061>
- Peters, S., Laanen, M., Groetsch, P., Ghezehegn, S., Poser, K., Hommersom, A., Pasterkamp, R., & Spaias, L. (2018). WISPstation: A new autonomous above water radiometer system. *Proceedings of the Ocean Optics XXIV Conference, Dubrovnik, Croatia*, 7–12.

- 275 <https://oceanopticsconference.org/proceedings/>
- 276 Rusch, T., & Mair, P. (2024). *cops: Cluster Optimized Proximity Scaling*. [https://doi.org/10.](https://doi.org/10.32614/CRAN.package.cops)
277 [32614/CRAN.package.cops](https://doi.org/10.32614/CRAN.package.cops)
- 278 Rusch, T., Mair, P., & Hornik, K. (2021). Cluster Optimized Proximity Scaling. *Journal*
279 *of Computational and Graphical Statistics*, 30(4), 1156–1167. [https://doi.org/10.1080/](https://doi.org/10.1080/10618600.2020.1869027)
280 [10618600.2020.1869027](https://doi.org/10.1080/10618600.2020.1869027)
- 281 Simis, S. G., Peters, S. W., & Gons, H. J. (2005). Remote sensing of the cyanobacterial
282 pigment phycocyanin in turbid inland water. *Limnology and Oceanography*, 50(1), 237–245.
283 <https://doi.org/10.4319/lo.2005.50.1.0237>
- 284 van der Woerd, H. J., & Wernand, M. R. (2015). True colour classification of natural waters
285 with medium-spectral resolution satellites: SeaWiFS, MODIS, MERIS and OLCI. *Sensors*,
286 15(10), 25663–25680. <https://doi.org/10.3390/s151025663>
- 287 van der Woerd, H. J., & Wernand, M. R. (2018). Hue-angle product for low to medium spatial
288 resolution optical satellite sensors. *Remote Sensing*, 10(2), 180. [https://doi.org/10.3390/](https://doi.org/10.3390/rs10020180)
289 [rs10020180](https://doi.org/10.3390/rs10020180)
- 290 Wei, J., Lee, Z., & Shang, S. (2016). A system to measure the data quality of spectral
291 remote-sensing reflectance of aquatic environments. *Journal of Geophysical Research:*
292 *Oceans*, 121(11), 8189–8207. <https://doi.org/10.1002/2016JC012126>
- 293 Zacharias, S., Lumpi, T., Weldon, J., Dirnböck, T., Gaillardet, J., Haase, P., Kühn, I.,
294 Vereecken, H., Bäck, J., Bergami, C., Bertsch-Höermann, B., Braud, I., Cools, N., Dick,
295 J., Dor-Haim, S., Forsius, M., Futter, M. N., Gaube, V., Groner, E., ... Mirtl, M. (2026).
296 Achieving harmonized and integrated long-term environmental observation of essential
297 ecosystem variables - the eLTER framework of standard observations. *Earth's Future*,
298 14(4), e2025EF006743. <https://doi.org/10.1029/2025EF006743>